

Summary Data on Acid Cropland and Lime Requirements in Malawi

GAIA

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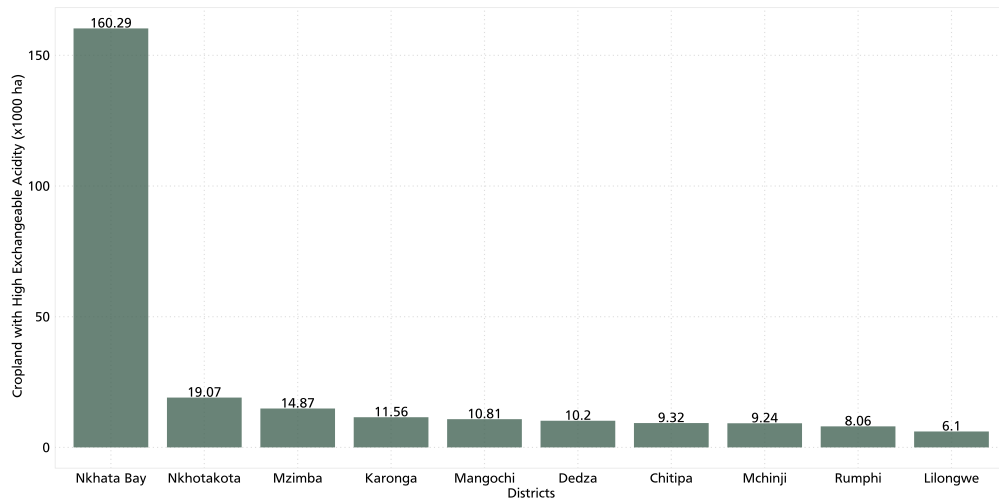
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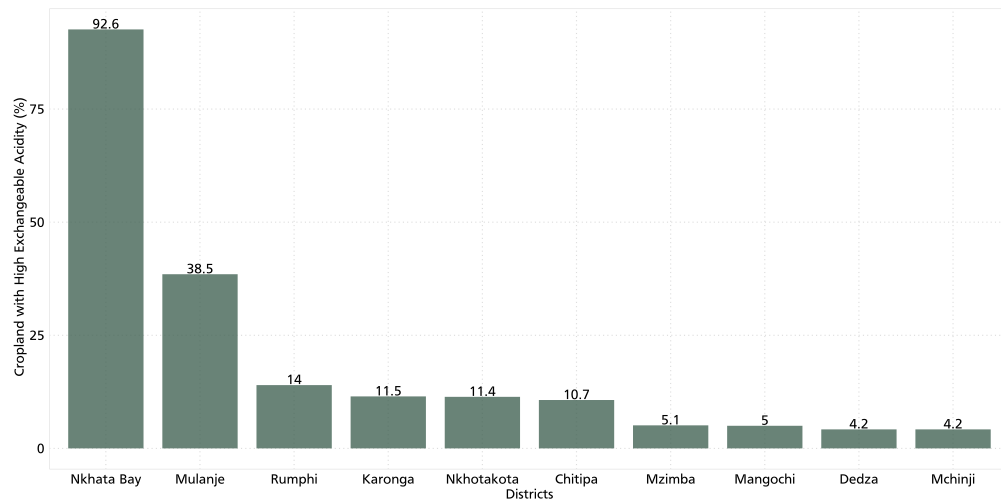
Introduction

- The slide deck provides a **data-driven, ex-ante analysis** of soil acidity across **Malawi**, evaluating the scale of affected areas, potential crop yield impacts, and the expected benefits of lime application for proactive soil remediation.
- **Highlights:**
 - **Acidic Cropland:** Identifies regions with high acidity levels impacting crop yields.
 - **Yield Loss:** Quantifies productivity losses in key crops due to acidic soils.
 - **Lime Requirements:** Estimates the lime amounts needed to improve soil health in these areas.
 - **Economic Potential:** Highlights estimated increases in production, ROI and profitability from lime application.
- Provide policymakers, farmers, and stakeholders with actionable insights to prioritize areas with high potential for productivity gains and economic return from soil remediation.

Top 10 Districts in Malawi with the highest acidic cropland



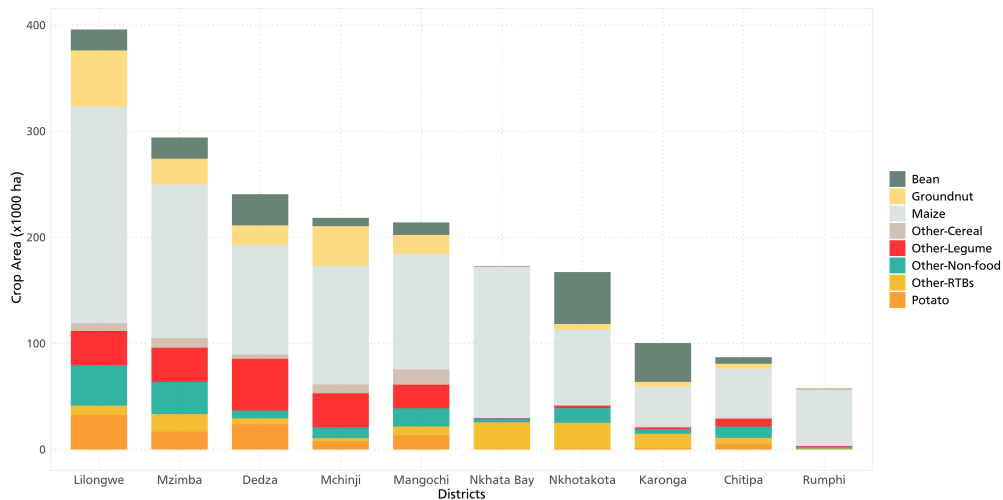
Top 10 Districts in Malawi with the highest acidic cropland (Percentage of total cropland)



└ Total crop area distribution (ha)

└ Total crop area in the top-10 most acidic Districts, not segregated by acidic cropland

Total crop area in the top-10 most acidic Districts, not segregated by acidic cropland



└ Total crop area distribution (ha)

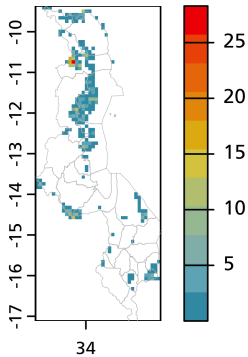
└ Total crop area in the top-10 most acidic Districts, not segregated by acidic cropland

Total crop area in the top-10 most acidic Districts, not segregated by acidic cropland (table)

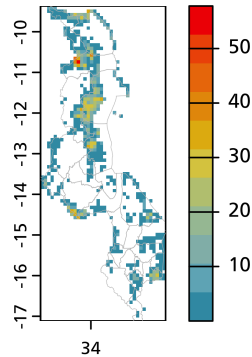
Districts	Bean	Groundnut	Maize	Other-Cereal	Other-Legume	Other-Non-food	Other-RTBs	Potato
Chitipa	6.05	4.73	46.09	1.22	7.25	10.64	5.47	5.50
Dedza	29.40	18.52	103.19	4.01	48.81	7.30	5.82	23.63
Karonga	36.59	4.24	38.04	0.80	1.45	4.36	13.71	1.16
Lilongwe	19.70	53.00	204.13	7.30	32.26	38.19	8.79	32.53
Mangochi	11.79	17.86	108.91	14.38	22.24	17.18	8.39	13.28
Mchinji	7.77	37.49	111.69	8.45	32.12	10.15	2.91	7.77
Mzimba	20.02	24.22	144.90	8.98	32.08	30.53	16.45	16.90
Nkhata Bay	0.48	0.17	142.57	0.12	0.28	3.75	25.48	0.17
Nkhotakota	49.01	5.37	71.21	0.25	2.43	13.76	24.52	0.69
Rumphi	0.62	0.67	52.61	0.31	0.76	1.57	0.55	0.34

Yield loss due to soil acidity for Maize and Groundnut in Malawi

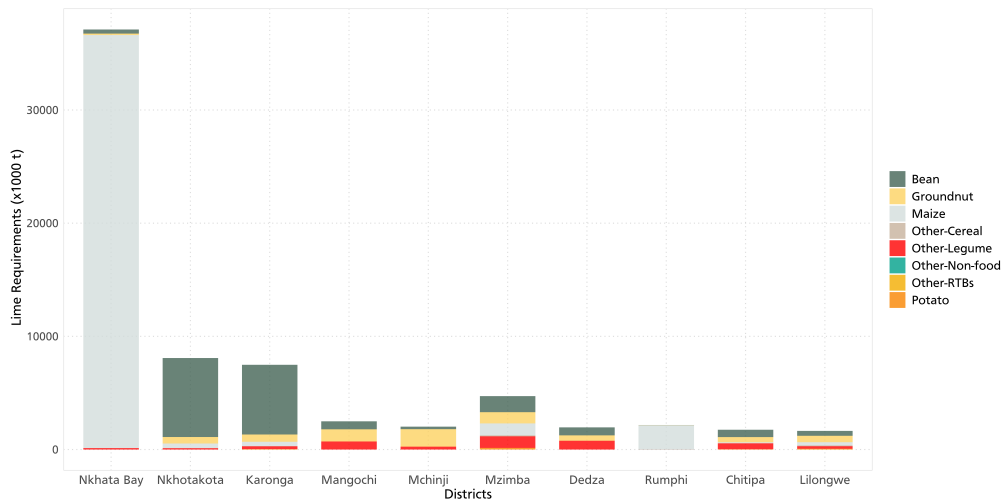
Maize Yield Loss (%)



Groundnut Yield Loss (%)



Lime requirements (t) for acid soil remediation in Malawi

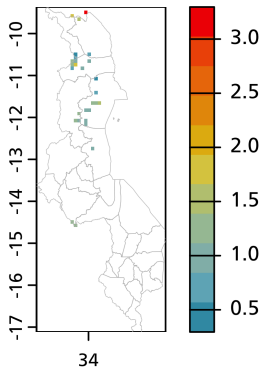


Lime requirements (1000 t) for acid soil remediation in Malawi (table)

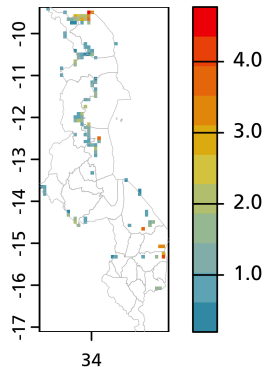
Districts	Bean	Groundnut	Maize	Other-Cereal	Other-Legume	Other-RTBs	Potato	Other-Non-food
Chitipa	0.65	0.45	0.09	0.01	0.53	-	0.02	-
Dedza	0.71	0.47	-	-	0.77	-	-	-
Karonga	6.16	0.64	0.38	0.02	0.25	-	0.04	-
Lilongwe	0.43	0.57	0.28	0.07	0.23	0.01	0.06	-
Mangochi	0.71	1.08	-	-	0.71	-	-	-
Mchinji	0.22	1.54	-	-	0.25	-	-	-
Mzimba	1.42	0.99	1.05	0.09	1.02	-	0.15	-
Nkhata Bay	0.37	0.13	36.50	-	0.12	-	-	-
Nkhotakota	6.99	0.57	0.41	-	0.12	-	-	-
Rumphi	0.01	0.01	2.12	0.01	0.01	-	-	0.01

Lime rate for Maize and Groundnut in Malawi

Maize Lime Rate (t/ha)

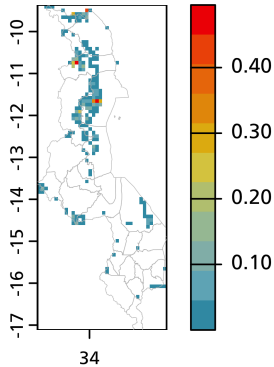


Groundnut Lime Rate (t/ha)

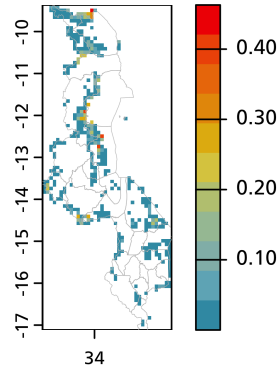


Yield response to lime application in Malawi

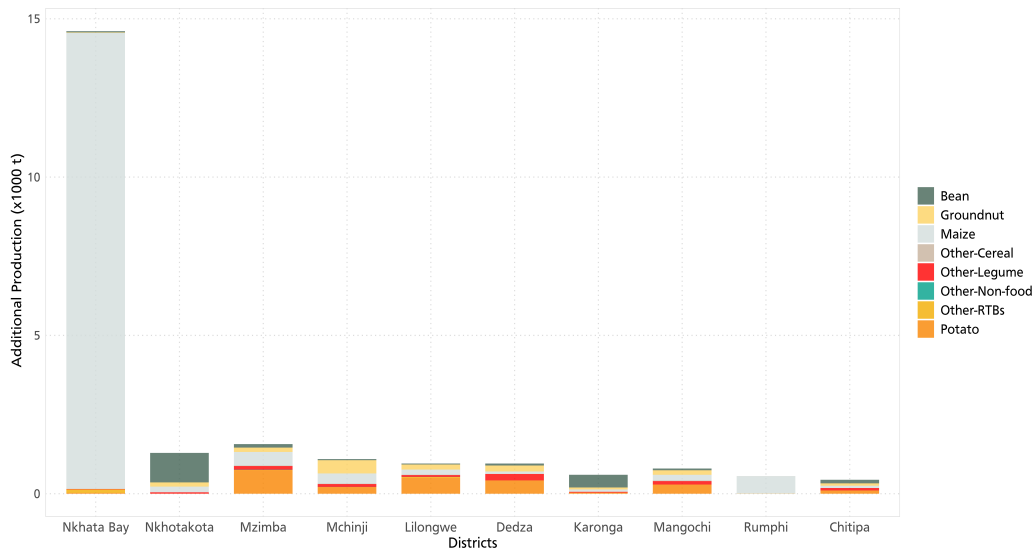
Maize Yield Response (t/ha)



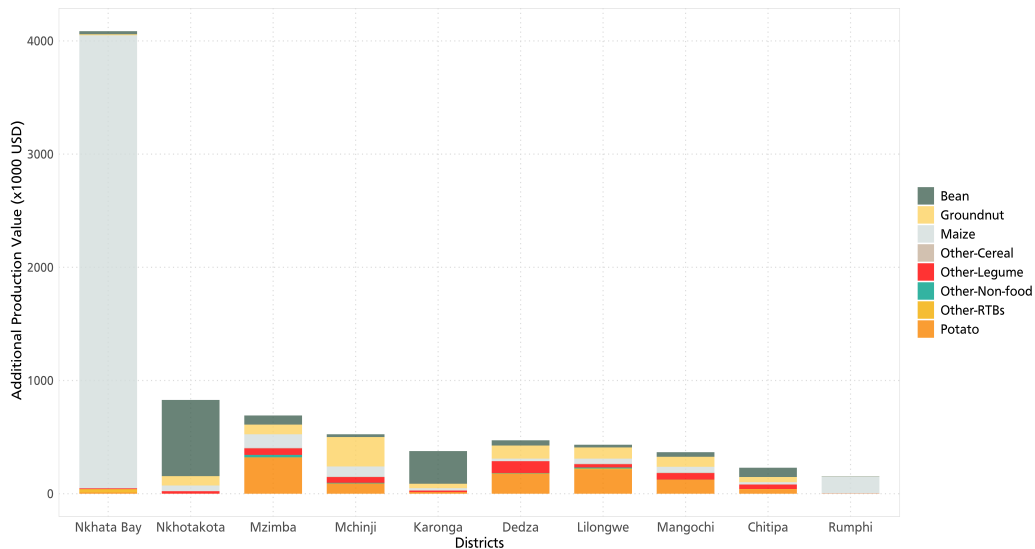
Groundnut Yield Response (t/ha)



Potential for additional production (tone) in top 10 acidic Districts in Malawi



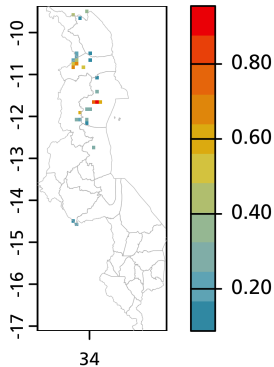
Potential for additional production value (US\$) in top 10 acidic Districts in Malawi



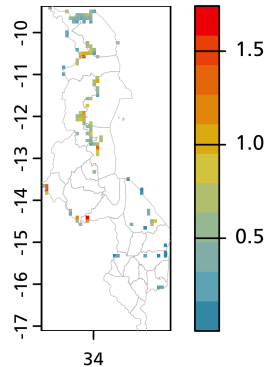
- ROI for Lime Application
- Return on investment for lime application in Malawi

Return on investment for lime application in Malawi

Maize ROI (USD/USD)

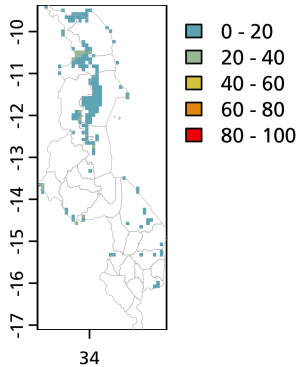


Groundnut ROI (USD/USD)

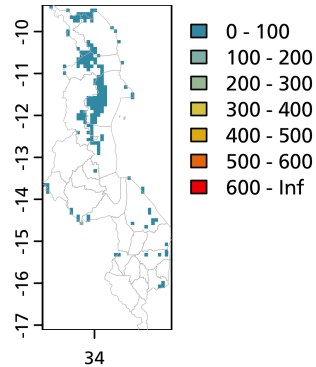


Profitability of lime application in Malawi

Profitable area (%)



Weighted profit (USD/ha)



Main assumptions and parameters

- Acidic cropland is defined as having acidity saturation levels above 10% of the Effective Cation Exchange Capacity (ECEC).
- Crop yields are unaffected by soil acidity until a specific acidity saturation threshold Target Acidity Saturation(TAS) is reached; beyond this threshold, yields decrease linearly, eventually reaching zero at high acidity levels.
- The TAS that separates maximum yield from yield decline is used as a critical value for making liming recommendations.
- Actual yields in acid tropical soils are a fraction of the potential yields in non-acidic soils, determined by the estimated yield loss due to acidity.
- Economic benefits from liming are calculated by multiplying the additional yield after soil remediation by median crop prices from sub-Saharan Africa (2016-2020 FAOSTAT data). The cost of liming is calculated using estimated lime application rates and a fixed lime price of 100 US\$/ton.
- The benefits of liming are assessed only for the year of application due to a lack of robust, large-scale data on long-term effects in sub-Saharan Africa.

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